BUSINESS ECONOMICS AND FINANCIAL ANALYSIS(21CS701HSM)

UNIT-3

TOPIC-1

Production Function

- Production function is the technical relationship between input and output.
- It shows how much output can be produced with a given set of inputs.
- The inputs for any product are Land, Labour, Capital,Organization and Technology.
- Mathematically the relationship between input and output can be expressed as –

$Q = f(L1, L2, C, O, T)$.

- Here Q is quantity of output, f is the function of , L1 is Land, L2 is Labour, C is capital, O is Organization, T is technology.

Production Function With One Variable Input / Laws of Returns

- The Laws of returns states that, when one factor is varied and all other factors are kept constant, then the total output increases in the initial stage at an increasing rate and after reaching a certain level of output it remains constant . If more and more doses of variable factor is added, the output will start declining.
- So this Law is also called as Law of Variable Proportions or Law of Diminishing returns.
- This law is of universal nature and proved to be true in agriculture and industry.

This Law can be illustrated with the help of following Table and Diagram.(Here Labour factor is varied and all other factors i.e Land, Capital, Technology etc. are kept constant.)

Units of Labour	Total Product(TP)	Marginal Product(MP)	Average Product(AP)
0	0	0	0 I STG
1	10	10	10
2	22	12	11 2STG
3	33	11	11
4	40	7	10
5	45	5	9
6	48	3	8 3 STG
7	48	0	7
8	45	-3	6

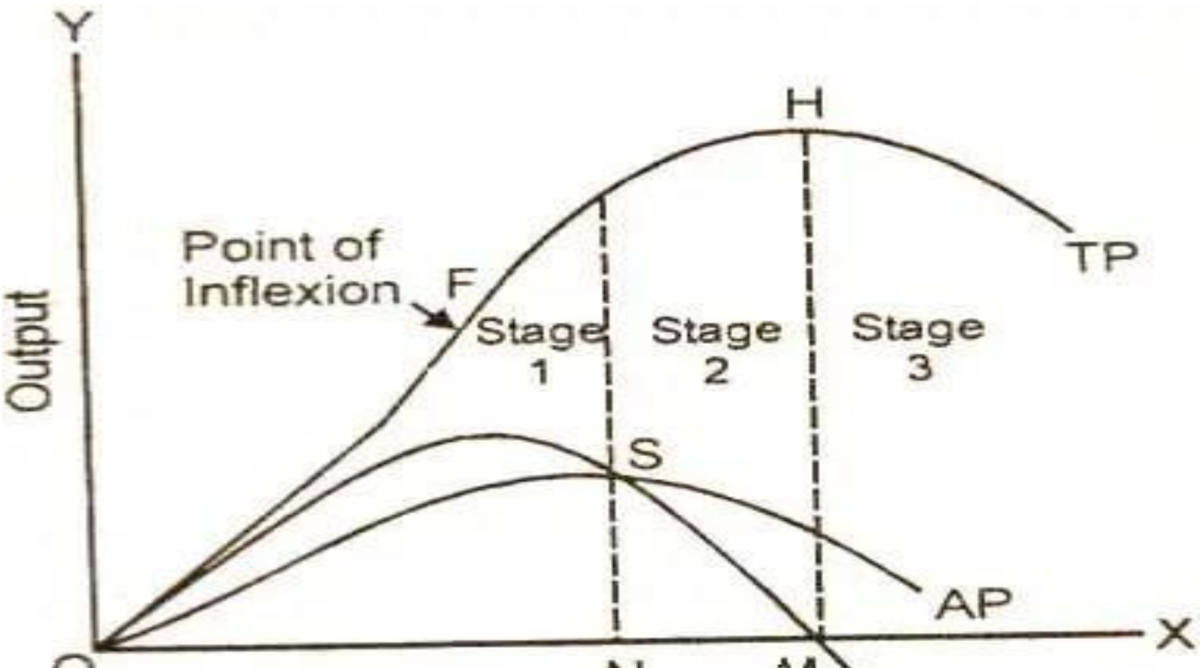


Fig. 11.1

- In the above diagram Units of labour represented on X-axis, output represented on Y-axis.
- TP represents Total Product. It is the sum of output produced by the variable factor. TP increases in the beginning, remains constant for a certain period, if more and more doses of labour is added , it starts declining.
- AP represents Average Product. Total Product divided by number of units gives AP. It also increases in the beginning (but at a slower rate than TP) remains constant, after certain period starts decreasing.
- MP represents Marginal Product. It is the change in output due to addition of one unit of variable factor. It raises faster and falls down faster.

✓ Stage I (Increasing Returns) :- In the above diagram TP Increases at a faster rate than AP and MP. At point "S" AP = MP . It indicates that TP will be levelled off even more and more doses of labour are added, and will not increase further. It indicates the end of Stage I.

- ✓ Stage 2 (Constant Returns) :- When TP is constant AP starts falling and MP falls more faster than AP. At point "M" MP cuts X-axis indicating the end of constant returns and on-setting of decreasing returns.

Stage 3 (Decreasing Returns) :- If further doses of labour are added the MP becomes negative and TP, AP will be showing decreasing returns

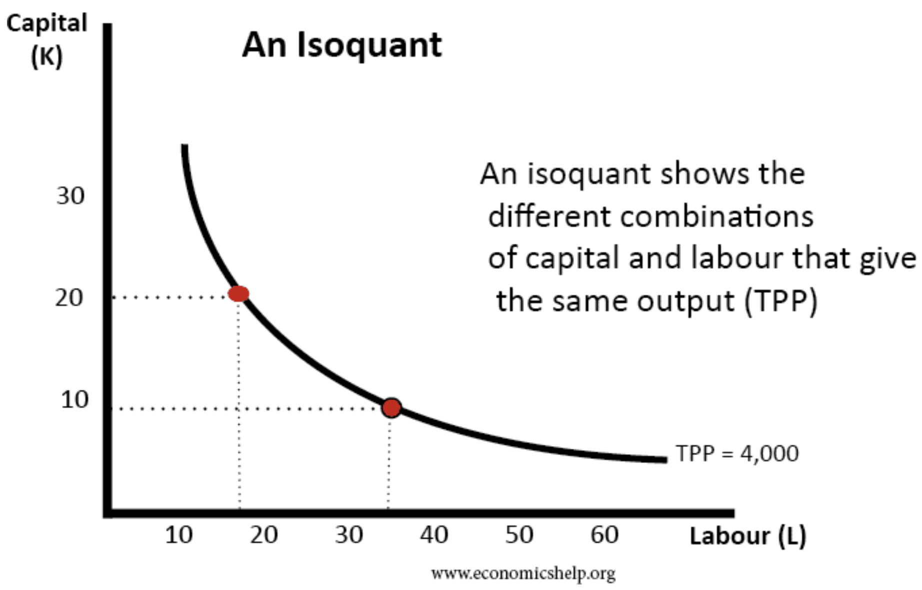
TOPIC-2

Production function with two variables

- In the previous concept , we have taken only Labour as variable factor.
- In this concept we will study Labour and Capital as variable factors.
- Labour and Capital are the two main inputs to produce a given quantity of output.
- Mathematically it can be represented as Q = f (C,L).
- These two inputs can be substituted to one another to produce a desired level of output.
- If one factor is increased , the other factor will be decreased and vice versa.
- More capital and less labour will be used for capital intensive products.
- More Labourand less Capital will be used for Labour intensive Products.
- It can be illustrated with the help of the Iso-Quant andIso- Cost curves.

Iso-Quant Curves:-

(Iso means equal , Quant means quantity) Iso-Quant curve shows that the level of output will be same at any point on the given iso-quant curve.It shows all the combination of two factors that produce a given output



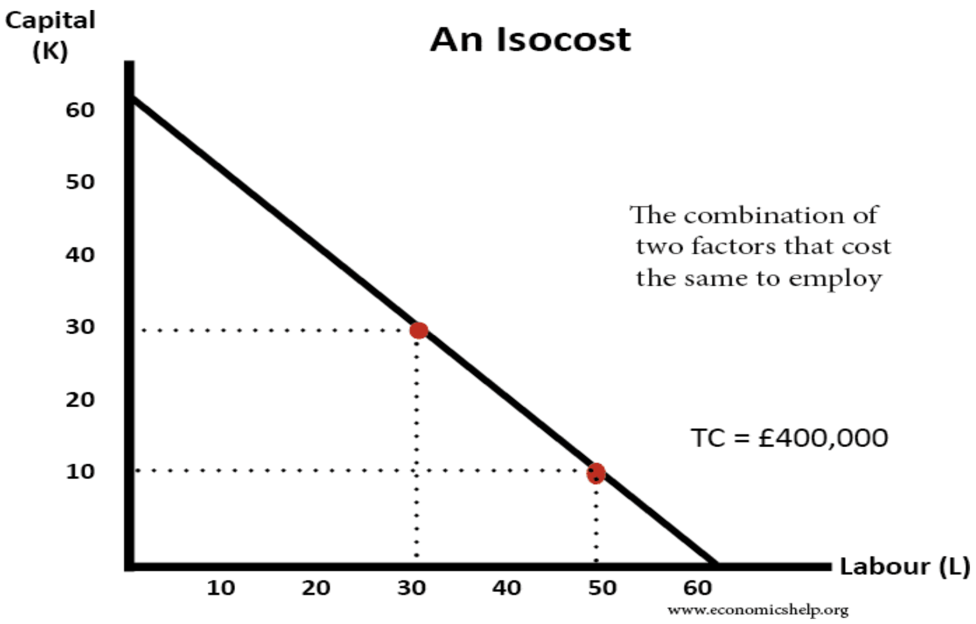
- ✓ In this diagram, the isoquant shows all the combinations of labour and capital that can produce a total output of 4,000 units.
- ✓ In this isoquant, this could be achieved through
- ✓ 20 capital and 18 labour

(or)

- ✓ 9 capital and 35 labour.
- ✓ The Iso-oquant curves are usually convex to the origin
- ✓ They will be parallel to one another.
- ✓ They will never touch x-axis and y-axis.
- ✓ If we want to increase the production the Iso-Quant curve will shift upwards (i.e to the right).

Iso-Cost Curves

Iso means equal . Cost means cost of production. Iso-cost curve represents the cost of production will be same at any point on the same Iso-cost curve.

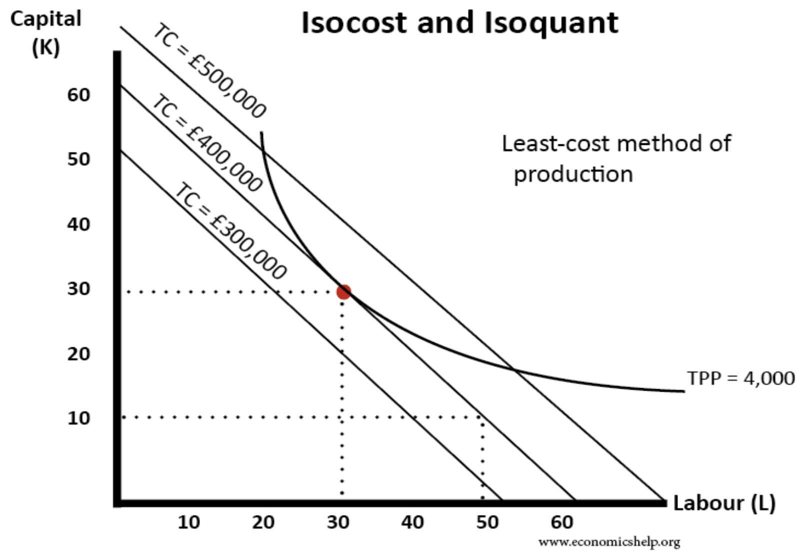


- ✓ An isocost shows all the combinations of factors that cost the same to employ.
- ✓ In this example, If we employ 30K and 30L, the total cost will be 4,00,000
- ✓ If we employ 10 K and 50L, the total cost will be 4,00,000
- ✓ The cost will be same at any point on the same curve
- ✓ If we want to increase the production , the curve will shift upwards to the right.

✓ They slope down wards from left to right. Parallel to one another.

<u>**Least cost combination of inputs**</u>.

- ❖ With the help of the Isoquant and Isocost curves the producer can find out optimum combination of inputs that costs him very less.
- ❖ If we super impose Isoquant curves on Isocost curves, the point where Isoquant curve is just tangential to the Isocostcuve determines the least combination of inputs.
- ❖ At this point the cost of production will be very less and helps the producer to maximise the profits. It is illustrated in the given diagram.



Long run production function/ Returns to scale

- In the long run all factors of production are variable. (I.e Land, Labour, Capital, Organisation, Technology.)
- It shows the relationship between input and output in the longrun.
- If we increase the inputs the output increases at a faster rate in the beginning, remains constant for a certain period and if more and more doses are added output starts decreasing.

TOPIC-3

Different types of Costs and their Behaviour

- ✓ Cost refers to the expenditure incurred to produce a particular product or service.
- ✓ Costs play a very important role in managerial decisions.
- ✓ Major part of product price consists of costs incurred for producing that product.
- ✓ Every producer tries to minimise costs to increase his profit margins.
- ✓ So it is important to understand various types of costs, their behaviour to control them effectively.

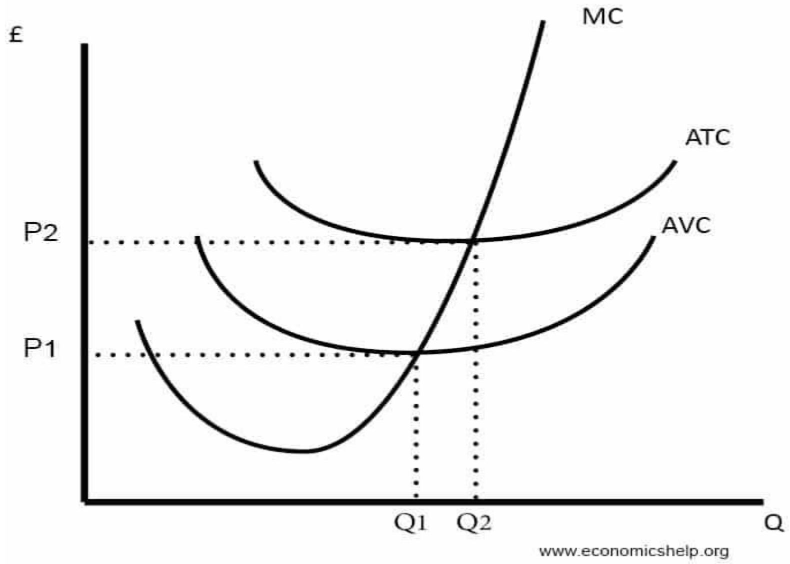
The various types of costs are as follows

- ✓ Long run vs Short run costs:- Long run is a period which is sufficient to change all factors of production. They are planning curves because management can plan its capacity of production. Short run is a time period during which only variable factors can be altered. Short run cost curves are called operating curves.
- ✓ Fixed vs Variable costs:- Fixed costs remain fixed irrespective of production. Where as variable costs vary with volume of production.
- ✓ Opportunity vs Actual cost:- Earnings that are foregone for losing an opportunity. Ex. Using own house for business and losing rent on it. Where as actual cost is one which is actually spent on a product.
- ✓ Incremental vs Sunk costs:- Incremental costs are the added costs due to increase in the level of production. Sunk costs are those which are already incurred and do not change with the level of producction.
- ✓ Explicit costs vs Implicit costs:- Explicit costs involve payment of cash. Ex. Rent, wages etc. Where as Implicit costs do not involve in payment of cash as they are not actually incurred. Ex. Saving in salary due to own supervision.

- ✓ <u>Out of pocket costs vs Book costs</u>:- Out of pocket costs are immediate cash flow in day-to-day business. Book costs are shown in the books and not required for current cash expenditure. Ex. Provision for depreciation, Bad debts.
- ✓ <u>Past costs vs Future costs</u>:- The costs which are already incurred are called past costs. They can not be controlled. Where as costs which will be incurred I future are called future costs and they can be contolled.
- ✓ <u>Urgent costs vs Postponable costs</u>:- Costs which affect the operational efficiency of the business and which must be met immediately are called urgent costs. Costs which are not urgent for continuation business are called postponable costs.

Short run cost curves and their behaviour

Short run is a time period during which , time is not sufficient to change Fixed costs like Land, Buildings, Machinary etc. So costs can be divided into Fixed and Variable costs. But in the long run all costs are variable . No distinction between Fixed and Variable costs.



- ✓ In this diagram, Average Variable costs (AVC) is a ‘U’ shaped curve denoting that, it tends to fall in the beginning when output is increasing, but after a particular level it will start increasing. AVC= Total VC / Q.
- ✓ Average Total Cost (ATC) is also ‘U’ shaped curve which will decrease in the beginning and raises after a certain period of time.

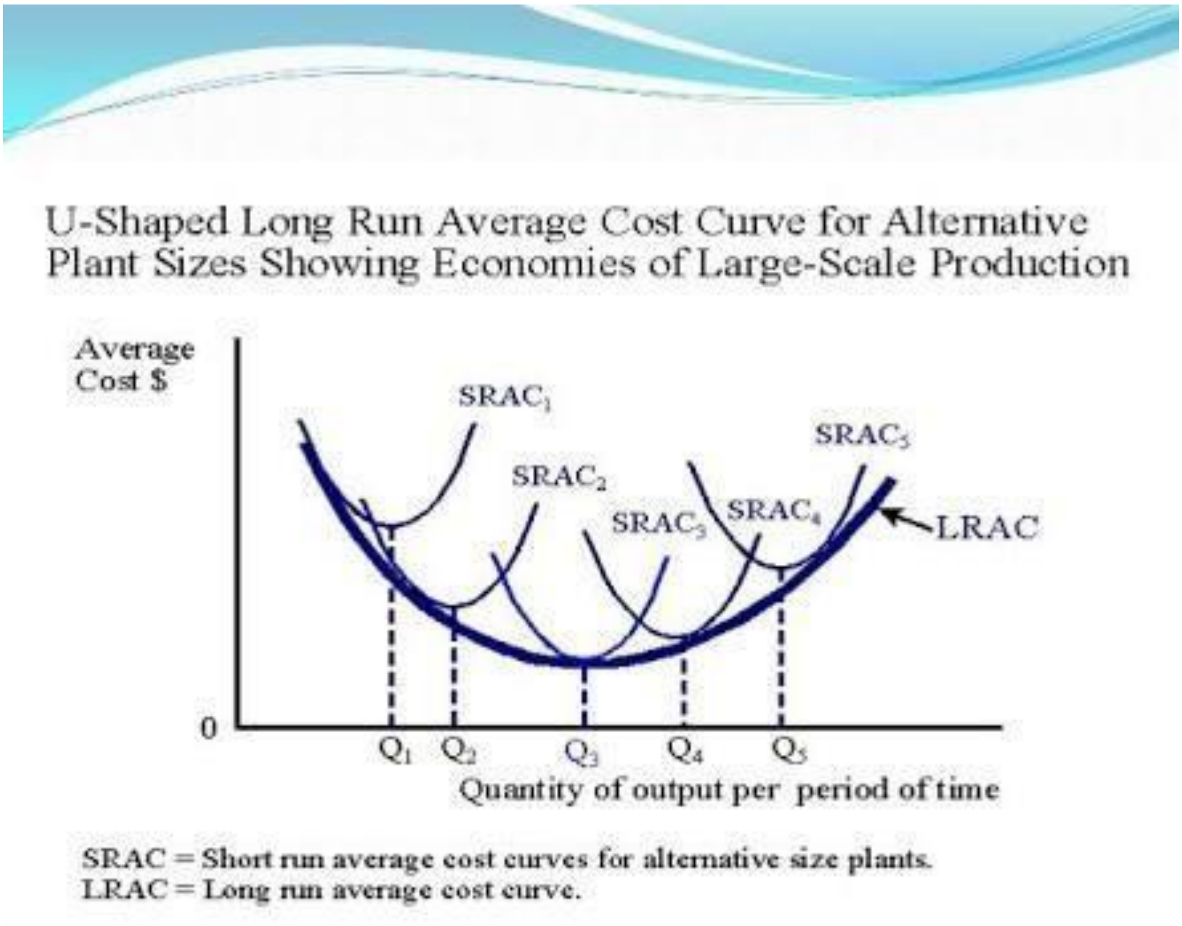
ATC = FC+VC / Q. or TC/Q

- ✓ Marginal cost (MC)is the additional cost incurred for producing one extra unit.It falls down fast and raises very fast.

MC = change in TC / Change in Q.

<u>Long run cost curves and their behavior</u>

- Long run refers to that time period during which firm can make changes in production, technology, can enter into new markets etc.
- All costs are variable.
- A long run is also expressed as a series of short run curves.



- In this diagram Long run average cost curve (LRAC) is a flat 'U' shaped curve which envelops a series of short run curves. i.e SRAC1, SRAC2 SRAC3 etc.

- Flat U shaped curve represents that the cost of production continues to be low till the output reaches OQ3. It is the optimum level bcz SRAC3 is tangential to the LARC. (SARC = LARC).

After this period if the firm produces more, cost of production starts raising,

indicated by OQ4 and OQ5 costs